LAB # 6

Yusuf Bozdağ 33830 Lab Section : C

1. Introduction

Function generators are widely used in electronics to produce different types of periodic signals such as square, triangular, and sinusoidal waveforms. These signals are important for testing and analyzing circuits, since many electronic systems respond differently depending on the input waveform. In this experiment, a function generator is designed using basic circuit components such as operational amplifiers, resistors, capacitors, and diodes.

The main goal of this lab is to build a system that can generate square, triangular, and sine-like waveforms with controllable frequency and amplitude. To achieve this, the circuit is divided into several stages, each performing a specific task. These stages are a Schmitt trigger, an integrator, and a waveform shaping circuit.

The Schmitt trigger is used to convert an input signal into a square wave by using positive feedback and hysteresis. This allows the circuit to switch cleanly between two voltage levels. The square wave is then fed into an integrator circuit, which converts it into a triangular waveform by integrating the signal over time. After that, a nonlinear shaping circuit made of diodes and resistors is used to modify the triangular waveform and make it closer to a sinusoidal shape. This is done by reducing the slope of the waveform gradually at higher voltage levels.

By connecting these stages together, a complete function generator is obtained. The system can continuously produce oscillations without needing an external periodic input. The frequency of the signal mainly depends on the integrator components, while the shape of the waveform is determined by the nonlinear behavior of the diode network. Overall, this experiment helps to understand how different circuit blocks can be combined to generate and shape signals in practical electronic systems.

2. Pre-Lab Work

In this experiment, a multi-stage signal generation and shaping system was designed using an operational amplifier. Starting from a triangular input signal, three main functional blocks were implemented: a Schmitt trigger, an integrator, and a sine wave converter.

First, the Schmitt trigger was used to convert the triangular waveform into a square wave by introducing hysteresis and improving switching behavior. Then, the integrator circuit was used to transform the square wave back into a triangular waveform by integrating the signal over time. Finally, a diode-based sine shaping network was employed to approximate a sinusoidal waveform from the triangular input.

Schmitt Trigger

The Schmitt trigger was implemented using an LM741 operational amplifier with positive feedback, as shown in the circuit schematic. The triangular input signal is applied through R3 = 3 kΩ, and part of the output is fed back through R1 = 15 kΩ to the same input node. According to the textbook, a positive-feedback bistable circuit has two stable states and switches when the input crosses certain threshold levels, creating hysteresis behavior.

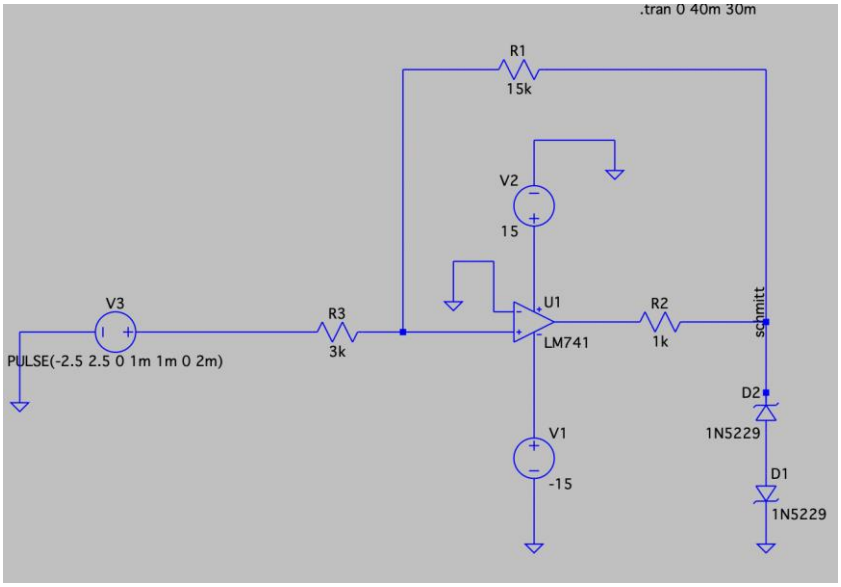


Figure 1.

The switching threshold is determined by the resistor division between R1 and R3. The threshold voltage is:

$$V_{\text{TH}} = \frac{R_3}{R_1 + R_3} \cdot V_{\text{out}}$$

Substituting the resistor values:

$$V_{\text{TH}} = \frac{3\ k\Omega}{15\ k\Omega + 3\ k\Omega} \cdot V_{\text{out}}$$

$$V_{\text{TH}} = \frac{3}{18} \cdot V_{\text{out}} = \frac{1}{6} \cdot V_{\text{out}}$$

Since the output is limited to approximately ±5 V by the Zener diodes, the threshold values become:

$$V_{\text{TH+}} = \frac{1}{6} \cdot 5 = 0.83\ V$$

$$V_{\text{TH-}} = \frac{1}{6} \cdot (-5) = -0.83\ V$$

These values were chosen so that the thresholds lie within the triangular input range (approximately ±2.5 V), as seen in the waveform figure. This ensures proper switching. The output waveform (green)

clearly switches between approximately ±5 V whenever the input crosses these threshold levels, confirming the expected hysteresis behavior.

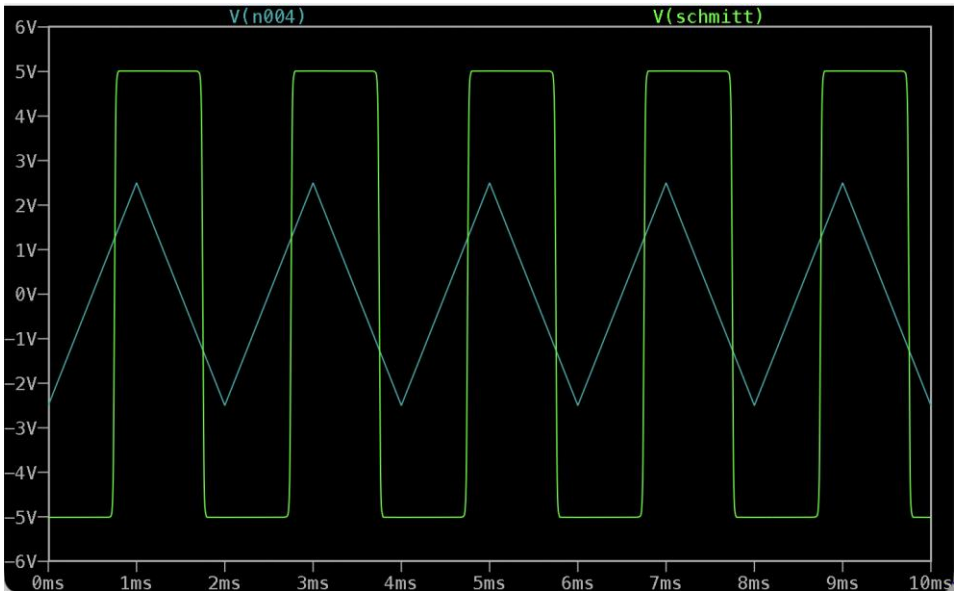


Figure 2.

Integrator

he integrator was implemented using an LM741 operational amplifier in an inverting configuration, as shown in the circuit. The square-wave output from the Schmitt trigger is applied through $R_1 = 10\ k\Omega$, and the capacitor $C_1 = 47\ nF$ is placed in the feedback path. This setup converts the square wave into a triangular wave by integrating the input signal over time.
To prevent output drift, a large resistor $R_2 = 100\ k\Omega$ is placed in parallel with the capacitor. This makes the circuit more stable in practice. The resistor $R_3 = 1\ k\Omega$ acts as a load.

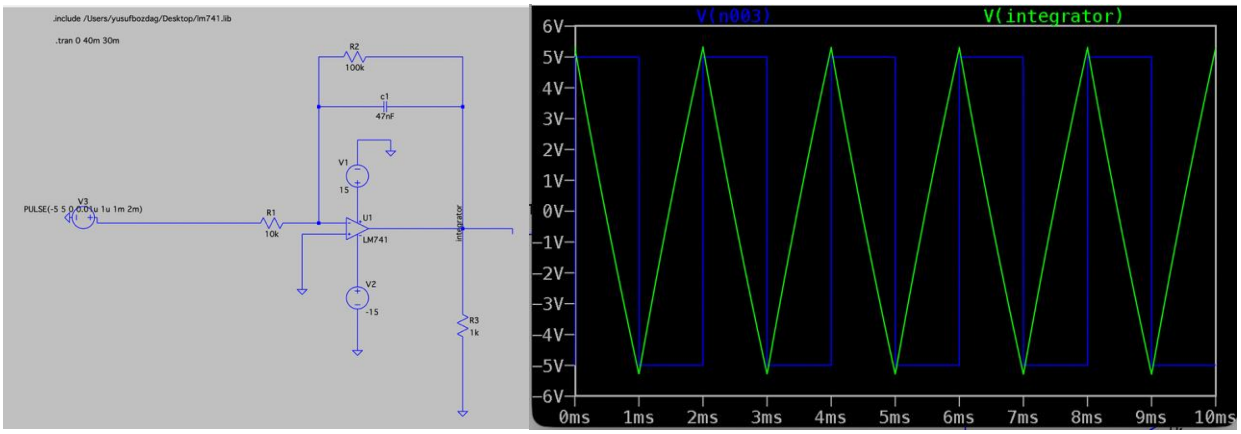


Figure 3 and 4.

The slope of the output is given by:

$$\frac{dV_{out}}{dt} = -\frac{V_{in}}{R_1 C_1}$$

First, calculate $R_1 C_1$:

$$R_1 C_1 = (10\ k\Omega)(47\ nF) = 470\ \mu s$$

For an input of $\pm 5\ V$:

$$\frac{dV_{out}}{dt} = \frac{5}{470\ \mu s} \approx 10.6\ V/ms$$

For **500 Hz**:

$$T = \frac{1}{500} = 2\ ms$$
$$\frac{T}{2} = 1\ ms$$

Voltage change:$\Delta V = 10.6 \times 1 = 10.6\ V$
So the output is approximately $\pm 5\ V$ triangular wave (matches the graph).

For **10 kHz**:

$$T = \frac{1}{10000} = 100\ \mu s$$
$$\frac{T}{2} = 50\ \mu s$$

Voltage change:

$$\Delta V = 10.6 \times 0.05 = 0.53\ V$$

So the output becomes much smaller at high frequency. The values R_1 and C_1 were chosen so that the integrator produces a full triangular waveform at low frequency (500 Hz). At higher frequencies (10 kHz), the amplitude decreases because the integrator has less time to charge and discharge.

Triangle to Sine Converter

The sine converter is implemented as a nonlinear shaping circuit using resistors and diodes, as shown in the figure. The triangular waveform is applied to the circuit, and the purpose is to gradually modify its slope to obtain a sine-like waveform.
When the input voltage is small, none of the diodes conduct. In this region, there is no current flowing through the shaping branches, and the circuit behaves linearly. Therefore, the output voltage is approximately equal to the input:

$$V_{out} = V_{in}$$

As the input voltage increases and exceeds a certain level V_1, one of the diodes starts conducting. At this point, the circuit behaves like a voltage divider, and the output becomes:

$$V_{out} = V_1 + (V_{in} - V_1)\frac{R_5}{R_4 + R_5}$$

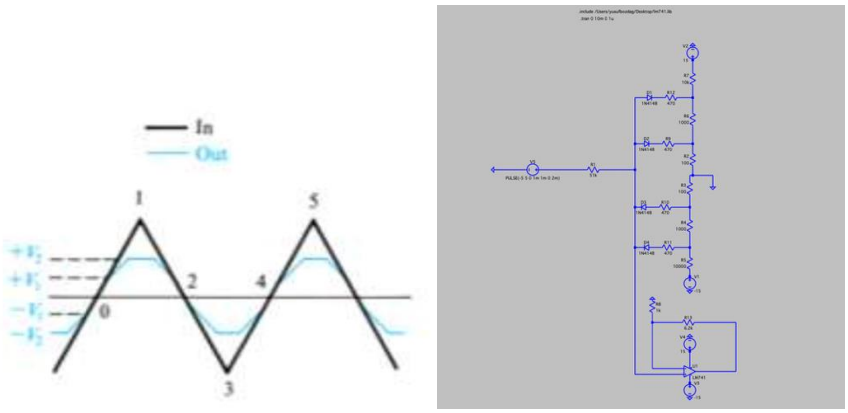


Figure 5 and 6.

The breakpoint voltages are generated by the symmetric resistor ladder on the right. The resistor values are chosen as $R_7 = R_5 = 10\,k\Omega$, $R_6 = R_4 = 1\,k\Omega$, and $R_2 = R_3 = 100\,\Omega$, connected between $+15\,V$ and $-15\,V$. Because of symmetry, the reference voltages are also symmetric around zero.
The current through the ladder is

$$I = \frac{15 - (-15)}{10k + 1k + 100 + 100 + 1k + 10k} = \frac{30}{22200} \approx 1.351\,mA$$

Voltage drops are

$$V_{10k} = I \cdot 10k \approx 13.51\,V, V_{1k} = I \cdot 1k \approx 1.351\,V, V_{100} = I \cdot 100 \approx 0.135\,V$$

Upper breakpoints:

$$V_2 = 15 - 13.51 = 1.49\,V, V_1 = 1.49 - 1.351 = 0.135\,V$$

By symmetry:

$$-V_1 = -0.135\,V, -V_2 = -1.49\,V$$

Including diode drop ($\approx 0.7\,V$), conduction starts approximately at

$$V_{B1} \approx 0.135 + 0.7 = 0.835\,V, V_{B2} \approx 1.49 + 0.7 = 2.19\,V$$

and similarly for negative values. Thus, for small inputs the output is linear, then the first diode reduces the slope after $0.835\,V$, and the second diode reduces it further after $2.19\,V$. This gradual slope reduction smooths the triangular waveform and produces a sine-like output.

This reduces the slope of the waveform. As the input increases further and reaches higher levels such as V_2, more diodes turn on, introducing additional slope reduction. In this way, the triangular waveform is divided into multiple regions, each with a smaller slope near the peaks. This piecewise-linear shaping smooths the sharp corners and creates an approximation of a sine wave.

The resistor values were selected to control both the breakpoints and the amount of slope reduction. The vertical resistor network sets the voltage levels V_1, V_2, and other transition points, while the resistors in series with the diodes determine how much the slope decreases in each region. These values were chosen symmetrically to obtain a balanced waveform around zero.

After the shaping stage, an op-amp stage is used to adjust the output amplitude. By selecting the feedback resistor appropriately, the signal is amplified to the desired voltage level without significantly distorting its shape. This ensures that the final output reaches the required amplitude while preserving the sine-like waveform.

In the waveform plot, the triangular input signal has a constant slope, while the shaped waveform shows reduced slope near the peaks. The final output is smoother and closer to a sine wave, and its amplitude is increased to the desired level by the op-amp stage.

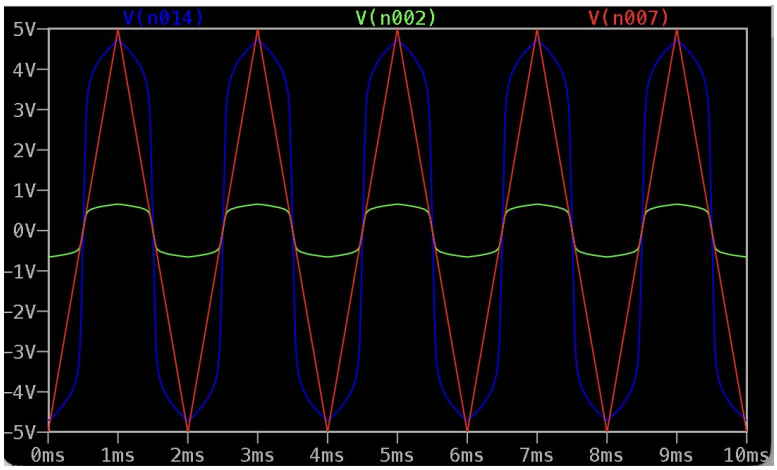


Figure 7.

Cascaaded System

The complete cascaded circuit consists of three main stages: a Schmitt trigger, an integrator, and a sine-shaping network followed by an amplification stage. These blocks are connected in a feedback loop to generate continuous oscillations without requiring an external periodic input.
The reason for closing the loop is to create a self-sustaining oscillator. The Schmitt trigger produces a square wave by switching between two voltage levels when its input crosses certain thresholds. This square wave is then applied to the integrator, which converts it into a triangular waveform by integrating the constant voltage levels over time. The triangular output is fed back into the Schmitt trigger input, completing the loop. As long as the triangular signal keeps crossing the threshold levels, the system continues to oscillate.

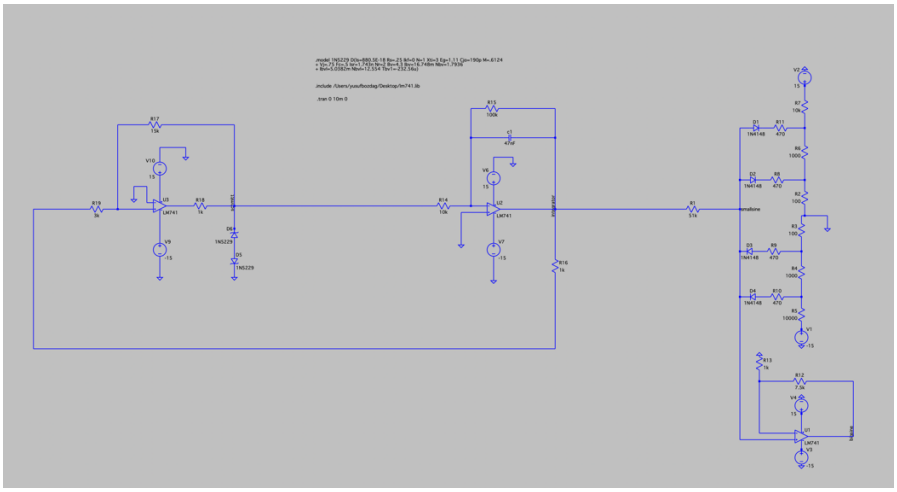


Figure 8.

In this configuration, each block has a clear role. The Schmitt trigger defines the switching thresholds and ensures stable transitions due to hysteresis. The integrator determines how fast the signal ramps up and down, which directly sets the oscillation frequency. The diode-based shaping network then modifies the triangular waveform into a sine-like waveform by reducing the slope in multiple regions. Finally, the last op-amp stage is used to scale the output to the desired voltage level without distorting the waveform.
The frequency of oscillation is mainly determined by the integrator stage. The slope of the integrator output is given by

$$\frac{dV_{out}}{dt} = -\frac{V_{in}}{RC}$$

and the oscillation frequency can be approximated by $f \approx \frac{1}{4RC}$

This shows that frequency is inversely proportional to both the resistor and capacitor values. In this circuit, the capacitor is fixed at $C = 47\,nF$

and the frequency is controlled by adjusting the integrator input resistor.
When $R = 10\,k\Omega$ the frequency becomes

$$f \approx \frac{1}{4(10\,k\Omega)(47\,nF)} \approx 500\,Hz$$

When the resistor is reduced to $R = 500\,\Omega$ the frequency increases to

$$f \approx \frac{1}{4(500)(47\,nF)} \approx 10\,kHz$$

Therefore, the frequency can be controlled by varying this resistor between approximately $10\,k\Omega$ and $500\,\Omega$. A larger resistance results in a lower frequency, while a smaller resistance produces a higher frequency.

Finally, the loop structure allows continuous oscillation, the Schmitt trigger provides clean switching, the integrator sets the frequency, and the nonlinear diode network shapes the waveform into a sine-like signal. The final op-amp stage ensures that the output reaches the desired amplitude

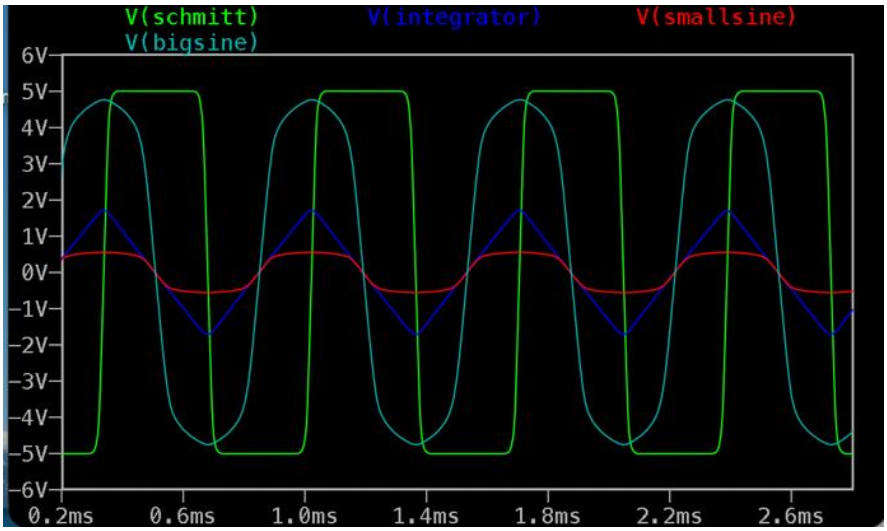


Figure 9.

3. Data/Results

3.1 Schmitt Trigger

Building Block

In this part of the experiment, a Schmitt trigger circuit was built in order to convert the triangular input waveform into a square wave. The Schmitt trigger is an op-amp based positive-feedback circuit that switches its output between two voltage levels when the input crosses certain threshold values. Because of this hysteresis behavior, the circuit produces a clean square waveform even when the input is a slowly changing ramp or triangular signal.

The circuit was implemented using one LM741 operational amplifier on the breadboard. The op-amp was placed across the center gap of the breadboard so that each pin could be connected easily to the surrounding rows. The non-inverting input of the op-amp was connected to ground, while the triangular input signal from the function generator was applied to the inverting input through a 3 kΩ resistor. In addition, a 15 kΩ resistor was connected from the output node back to the same inverting input node to provide positive feedback. This feedback path is the key part of the Schmitt trigger operation, since it creates two different switching thresholds and causes the output to change state only when the input crosses these levels.

At the output side, a 1 kΩ resistor was connected in series with the op-amp output. After this resistor, two 1N5229 Zener diodes were connected in opposite directions to ground. This part of the circuit was used to limit the output voltage and keep the square-wave amplitude around a fixed value instead of letting the op-amp saturate fully at the supply rails. The op-amp itself was powered by +15 V and -15 V supply connections through the breadboard power rails. In the breadboard implementation, jumper wires were used to bring the supply voltages, input signal, and ground to the correct rows, while the resistors and Zener diodes were connected directly between the required nodes to match the schematic in Figure 1.

Components used:
LM741 op-amp
R1 = 15 kΩ, R2 = 1 kΩ, R3 = 3 kΩ
2 × 1N5229 Zener diodes
Breadboard
Function generator
Oscilloscope
Connecting wires and probes
+15 V and -15 V DC supply

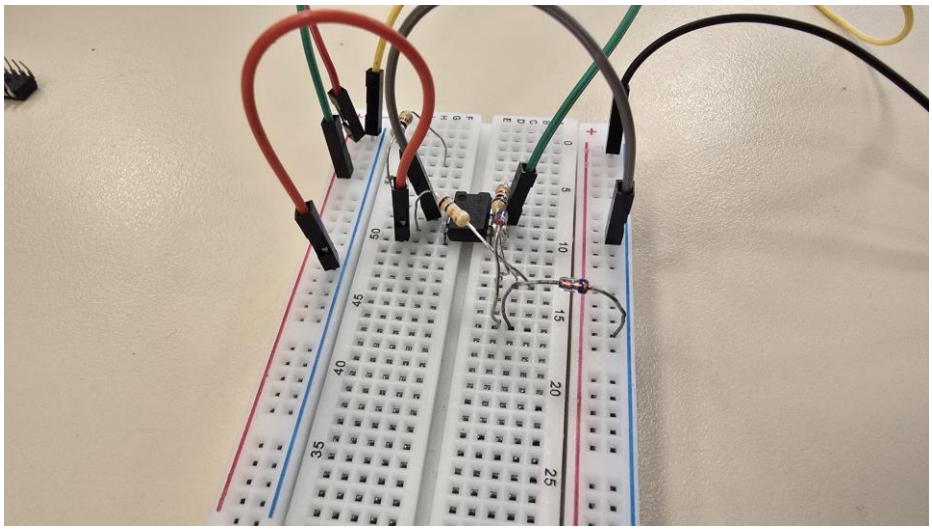


Figure 9. Breadboard implementation of the Schmitt trigger circuit.

Figure 9 shows the breadboard implementation of the Schmitt trigger circuit. The LM741 is placed at the center of the breadboard, and the surrounding resistors and Zener diodes form the input, feedback, and output-limiting network.

The circuit was tested by applying a 5 Vpp ramp input from the function generator. The output waveform was then observed on the oscilloscope. At 500 Hz, the measured output was a square wave with approximately 5.3 V peak-to-peak amplitude. At 10 kHz, the output waveform remained square and stable, with approximately 5.4 V peak-to-peak amplitude. These measurements show that the circuit successfully converted the ramp input into a square wave at both low and high operating frequencies.

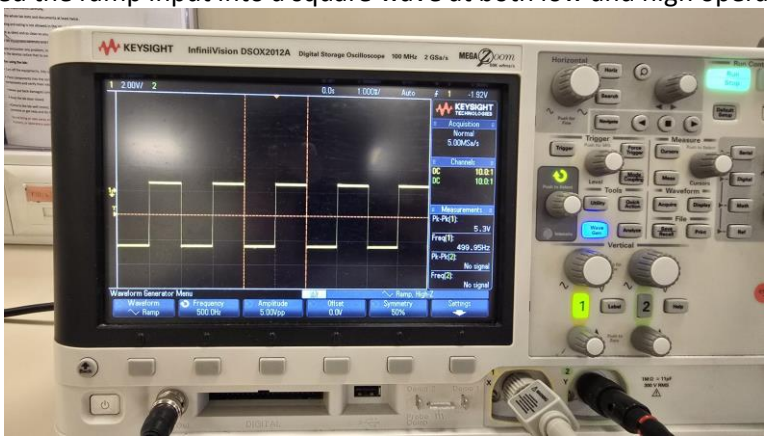


Figure 10. Oscilloscope output of the Schmitt trigger at 500 Hz.



Figure 11. Oscilloscope output of the Schmitt trigger at 10 kHz.

The measured results confirm the expected behavior of the Schmitt trigger. As the ramp input increased and decreased, the output switched sharply between two voltage levels, producing a symmetrical square waveform. The Zener diodes effectively limited the output amplitude to nearly 5 V, and the circuit maintained proper switching operation at both 500 Hz and 10 kHz. Therefore, the experimental results agree with the theoretical operation of the Schmitt trigger and show that this block can be used successfully as the first stage of the function generator.

3.2 Integrator

Building Block

In this part of the experiment, an integrator circuit was implemented to convert the square wave signal into a triangular waveform. The integrator is an op-amp based circuit that produces an output proportional to the integral of the input signal over time. Therefore, when a square wave is applied, the output becomes a linearly increasing and decreasing signal, which forms a triangular waveform.
The circuit was built using an LM741 operational amplifier placed at the center of the breadboard. The non-inverting input of the op-amp was connected to ground. The square wave signal coming from the Schmitt trigger stage was applied to the inverting input through a resistor. In the feedback path, a capacitor was connected between the output and the inverting input to perform the integration operation. In addition, a large resistor was placed in parallel with the capacitor to improve stability and prevent output drift during operation.

In the design of the integrator, the values of the resistor and capacitor were selected based on the theoretical relationship; dVout/dt = -Vin / (RC). Using the pre-lab calculations (figure 3), the capacitor value was fixed, and the resistor value was adjusted manually by changing resistor values on the breadboard instead of using a potentiometer. This allowed the RC time constant to be tuned so that the desired triangular waveform amplitude was obtained. In this experiment, C was kept constant while different resistor values were tested until the output matched the expected behavior.

Components used:

LM741 op-amp
R ≈ 10 kΩ and 510Ω (adjusted manually)
C = 47 nF
Additional resistor (for stability)
Breadboard, wires, oscilloscope probes

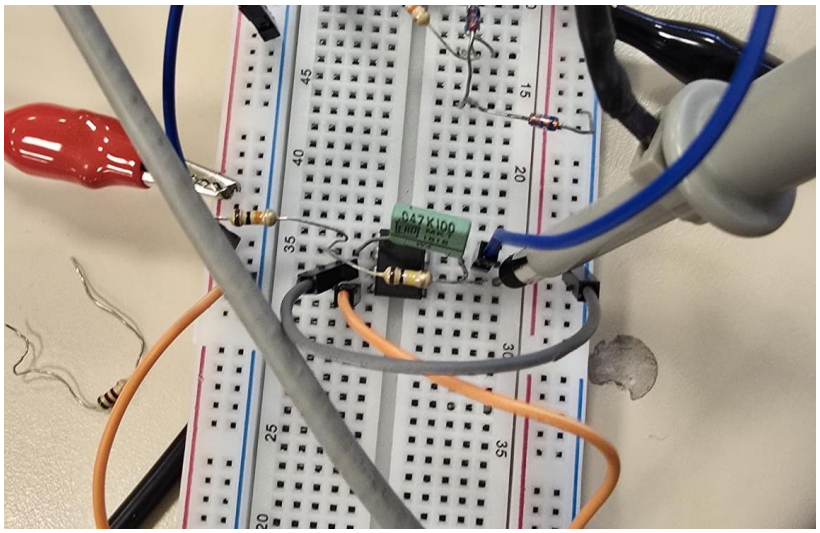


Figure 12. Breadboard implementation of the integrator circuit.

The circuit was tested using square wave inputs at both 500 Hz and 10 kHz. At 500 Hz, the output waveform was a clean and symmetric triangular wave with approximately 5 V peak amplitude, which matches the expected theoretical result. The slope of the waveform was consistent with the RC value calculated in the pre-lab.

The capacitor value was fixed at 47 nF, and the resistor was adjusted manually to set the operating frequency. In this implementation, a 10 kΩ resistor was used for the lower frequency case (around 500 Hz–5 kHz), while a 510 Ω resistor was used for the higher frequency case (around 10 kHz). Instead of using a potentiometer, standard resistor values were selected and replaced directly on the breadboard to achieve the desired RC time constant.

At 10 kHz, the output waveform started to deviate from an ideal triangular shape. Distortions appeared especially near the corner points of the waveform, where sharp transitions and small spikes were observed. Because of these distortions on figure 14, the measured peak-to-peak voltage appeared slightly higher than 5 V. However, when focusing on the main triangular shape of the waveform, the amplitude of the signal was clearly around 5 V, as expected.

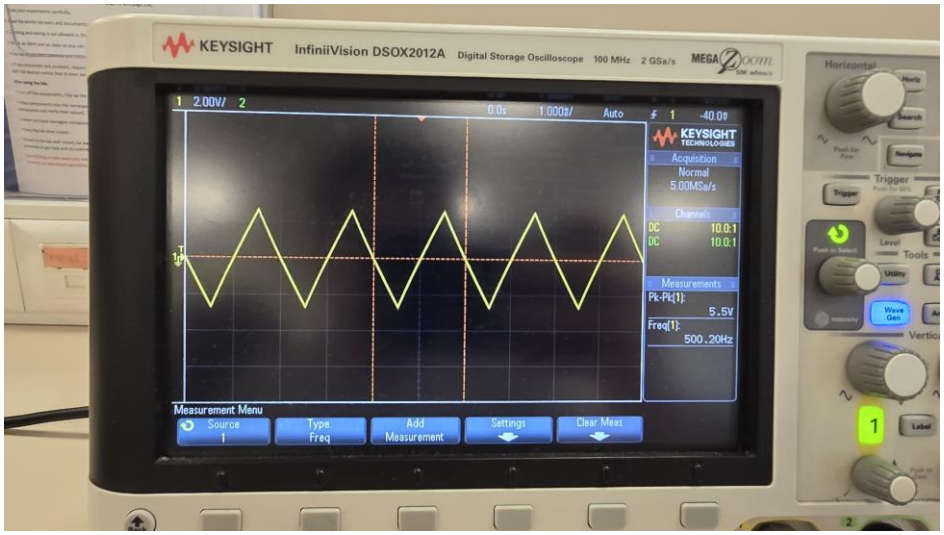


Figure 13. Oscilloscope output of the integrator at 500 Hz.

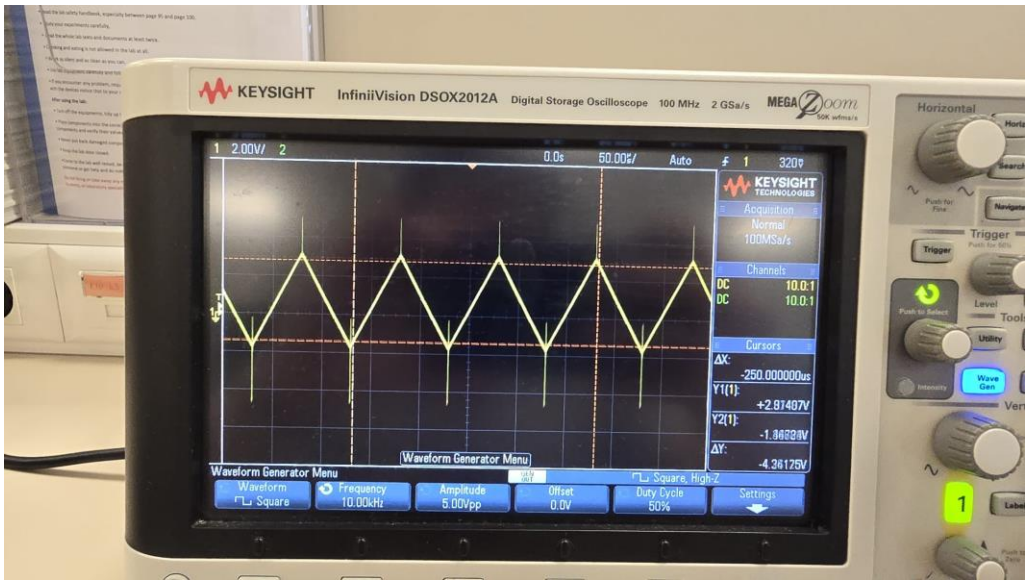


Figure 14. Oscilloscope output of the integrator at 10 kHz.

Overall, the integrator successfully converted the square wave into a triangular waveform. The results are consistent with the theoretical expectations at low frequency, while at higher frequencies, non-ideal effects such as limited op-amp response and sharp switching transitions caused slight distortions in the output signal.

3.3 Triangle to Sine Wave Generator

Building Block

In this part of the experiment, a triangular-to-sine wave generator was designed and tested as an independent block rather than as part of the full cascaded oscillator. Therefore, the input waveform was not taken from the integrator stage of the complete system. Instead, a triangular input signal was directly applied from the function generator, and the response of the sine-shaping circuit was observed on the oscilloscope. This approach made it possible to study the behavior of the shaping circuit alone and to verify whether it could successfully transform a triangular waveform into a sine-like waveform under the required operating conditions.

The purpose of this block is to approximate a sinusoidal waveform by reshaping a triangular input through a nonlinear resistor-diode network. In an ideal triangular wave, the slope is constant during both the rising and falling portions of the signal. A sine wave, however, does not have a constant slope. Its slope is highest near the zero-crossing region and becomes smaller as the waveform approaches its positive and negative peaks. For this reason, if a triangular waveform is to be converted into a sine-like waveform, the sharp corners must be softened and the slope near the peaks must be reduced gradually. The circuit used in this experiment performs exactly this operation by using several diode branches that begin to conduct at different input voltage levels. As the input amplitude increases, more branches become active, and the effective transfer characteristic changes step by step. This creates a piecewise-linear approximation of a sine wave rather than a simple clipped waveform

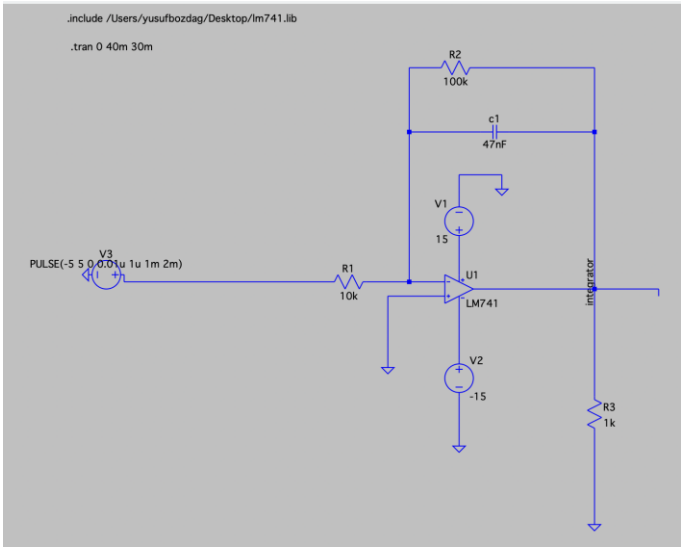
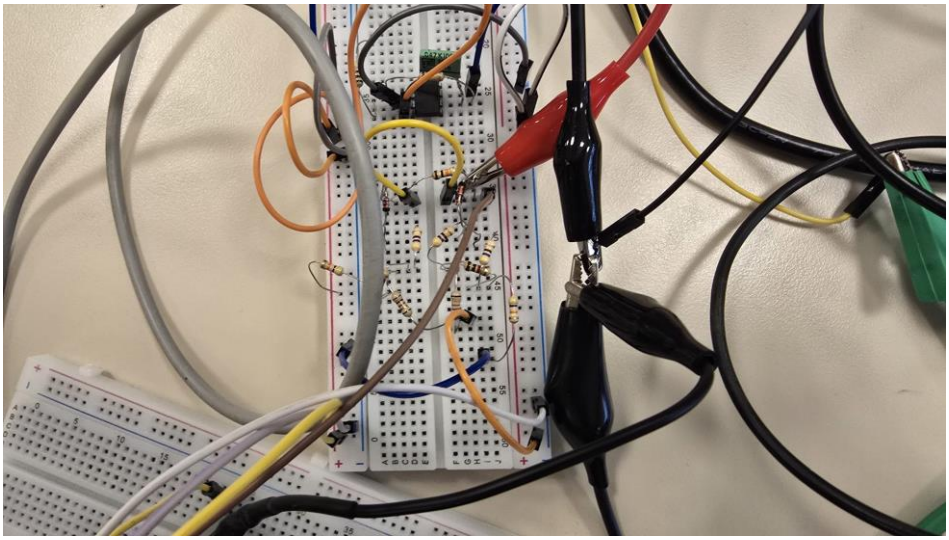


Figure 15 schematic of implemented sine wave generator

The implementation started with the resistor-diode shaping network shown in the pre-lab design. The triangular input from the function generator was first applied through the input resistor R1 = 51 kΩ. This resistor limits the input current and also participates in determining how strongly the shaping branches affect the waveform. From the main signal node, several diode-resistor branches were connected to a symmetric resistor ladder biased between +15 V and -15 V. In the implemented circuit, the ladder resistors were chosen as R7 = 10 kΩ, R6 = 1 kΩ, R2 = 100 Ω, R3 = 100 Ω, R4 = 1 kΩ, and R5 = 10 kΩ, connected in series between the positive and negative supply rails. Because the network is symmetric around the center node, the reference voltages generated by the ladder are also symmetric with respect to ground. This symmetry is important because it ensures that the positive and negative halves of the output waveform are shaped in the same way. Without this symmetry, the output could become distorted or offset to one side.


Figure 16. Breadboard implementation

In addition to the ladder network, each shaping branch included a diode and a resistor. In the implemented circuit, the diodes were 1N4148 diodes, and the branch resistors were chosen as R9 = R10 = R11 = R12 = 470 Ω. These branches were connected from the main signal node to different points of the resistor ladder. Their role was to activate at different signal levels and progressively alter the slope of the waveform. When the input voltage is small, the diodes are off, so almost no current flows through these branches. In that case, the shaping network has little effect and the output approximately follows the triangular input. As the input grows and reaches the first breakpoint, one pair of diodes begins to conduct. This creates an additional current path and changes the effective resistance seen by the signal. As the input increases further and reaches the next breakpoint, more branches begin to conduct, reducing the slope even more. In this way, the waveform is not abruptly clipped, but smoothly compressed near the peaks, which produces the rounded upper and lower portions characteristic of a sine-like signal

The resistor values used in the shaping ladder were not chosen randomly. In the pre-lab, these values were selected so that the circuit would generate suitable breakpoint voltages. The total voltage across the ladder is 30 V, since the network is connected between +15 V and -15 V. The total resistance of the ladder is 10 kΩ + 1 kΩ + 100 Ω + 100 Ω + 1 kΩ + 10 kΩ = 22.2 kΩ Using Ohm's law, the current through the ladder is I = 30 / 22200 ≈ 1.351 mA

Once this ladder current is known, the voltage drops across the individual resistors can be calculated. The drop across each 10 kΩ resistor is approximately V10k = (1.351 mA)(10 kΩ) ≈ 13.51 V The drop across each 1 kΩ resistor is approximately V1k = (1.351 mA)(1 kΩ) ≈ 1.351 V. The drop across each 100 Ω resistor is approximately V100 = (1.351 mA)(100 Ω) ≈ 0.135 V

From these drops, the breakpoint reference voltages were determined in the pre-lab as approximately V2 ≈ 15 - 13.51 = 1.49 V V1 ≈ 1.49 - 1.351 = 0.135 V and by symmetry. -V1 ≈ -0.135 V. -V2 ≈ -1.49 V

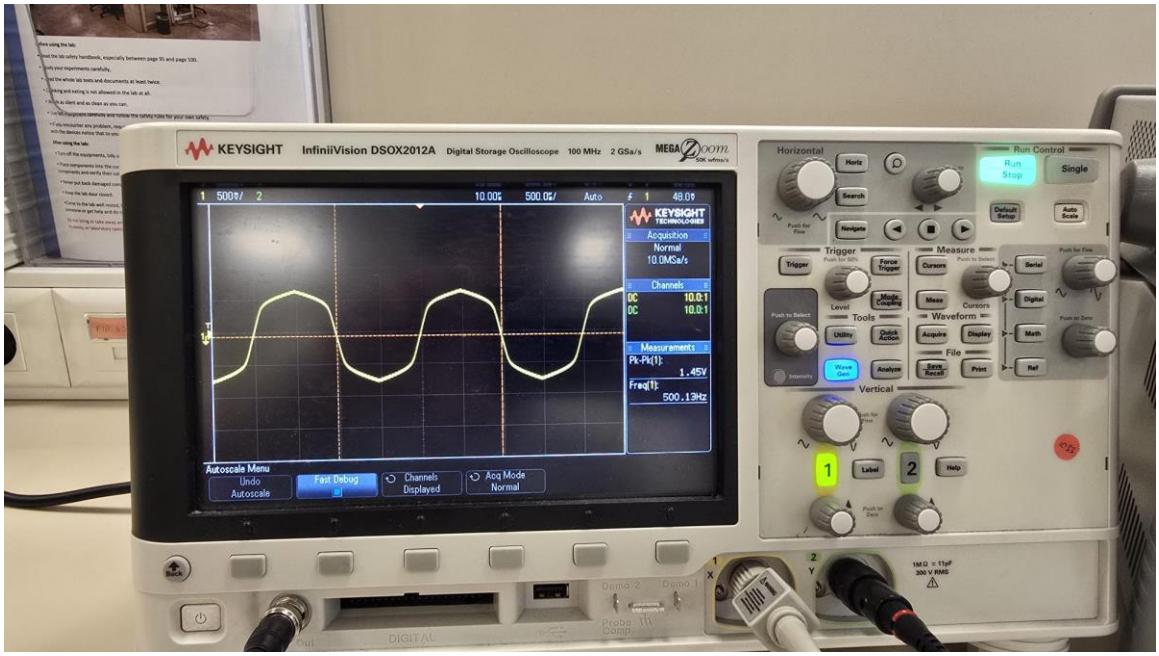


Figure 17.

Since the shaping branches contain silicon diodes, the forward voltage drop of each diode must also be considered. Taking the diode drop as about 0.7 V, the effective conduction thresholds become VB1 ≈ 0.135 + 0.7 = 0.835 V. VB2 ≈ 1.49 + 0.7 = 2.19 V with corresponding negative breakpoints on the opposite side. These results explain the operating principle of the circuit: for small input values, the output is nearly linear, then after about 0.835 V the first shaping branch begins to conduct and reduces the slope, and after about 2.19 V the next branch also conducts and reduces the slope further. This is the key reason why the output becomes rounded rather than triangular

After implementing the shaping circuit, its output was measured directly on the oscilloscope. The triangular input signal was applied from the function generator at both 500 Hz and 10 kHz, since these are the main frequencies required by the lab. During the first measurements, the output of the shaping circuit was found to have an amplitude of approximately 1.45 V. This result is consistent with the role of the shaping network: it does not amplify the signal, but rather reshapes and partially compresses it. Therefore, an additional gain stage was needed to satisfy the lab requirement of obtaining 5 V and 10 V sine-like outputs.

For this reason, a second stage was implemented using an LM741 operational amplifier as a gain amplifier. In the implemented amplifier stage, the resistor values were R8 = 1 kΩ and R13 = 6.2 kΩ. These values determine the voltage gain of the amplifier. Based on the resistor ratio, the gain is approximately set by the feedback-to-input resistor ratio. Since the measured shaped output was around 1.45 V, the required gain for a 10 V output is approximately Gain ≈ 10 / 1.45 ≈ 6.9

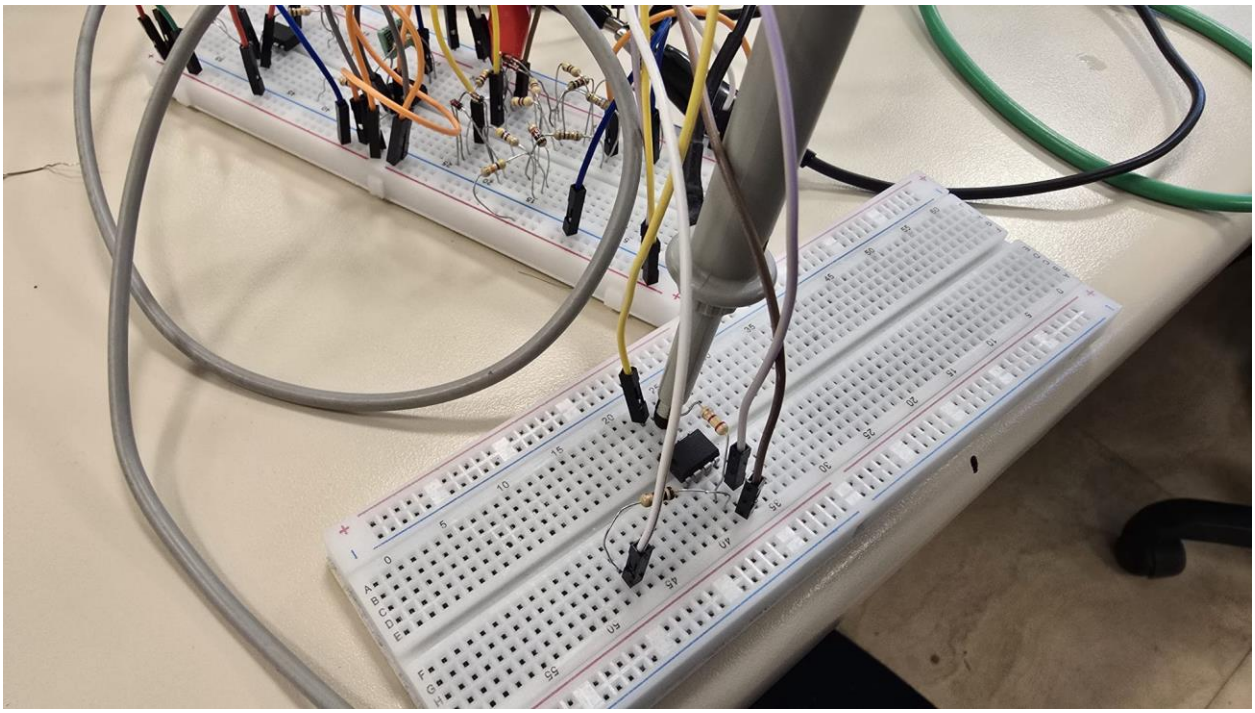


Figure 18. Implementation of non inverting opamp.

The selected resistor pair of 6.2 kΩ and 1 kΩ gives a gain close to this value in practical operation, allowing the signal to be increased to around 10 V without strongly distorting its shape. Similarly, for the 5 V output case, the same signal can be amplified with a lower gain setting, or the measured output can

be adjusted so that the final amplitude remains near 5 V. Thus, the values used in the op-amp stage were chosen according to the ratio between the measured amplitude after the shaping network and the target output amplitude required by the lab.

The physical implementation on the breadboard was done in two main sections. One section contained the resistor-diode shaping network, where the 1N4148 diodes and the ladder resistors were arranged symmetrically between the ±15 V rails. Care was taken to connect the diode directions correctly, because the positive and negative halves of the shaping process depend on the orientation of the diodes. The second section contained the LM741 amplifier stage. The op-amp was placed across the center gap of the breadboard, and its supply pins were connected to +15 V and -15 V. The input from the shaping network was then connected to the op-amp stage, and the final amplified output was observed on the oscilloscope. In this way, the triangular-to-sine conversion and the amplitude adjustment were tested as two consecutive but separately understandable operations.

The experimental results support the expected operation of the circuit. At 500 Hz, the waveform at the output of the shaping circuit was visibly smoother than the original triangular input. The straight-line segments of the triangle became rounded, especially near the top and bottom of the signal, indicating that the diode branches were activating and reducing the slope in those regions. After amplification, the output amplitude could be increased to approximately 5 V and 10 V as required. At 10 kHz, the waveform still maintained a sine-like appearance, although the output was somewhat less ideal than at 500 Hz. This behavior is reasonable because at higher frequencies the non-ideal characteristics of the diodes and the finite bandwidth and slew-rate limitations of the op-amp become more significant. Even so, the output remained clearly more sinusoidal than triangular, showing that the shaping network continued to function properly.

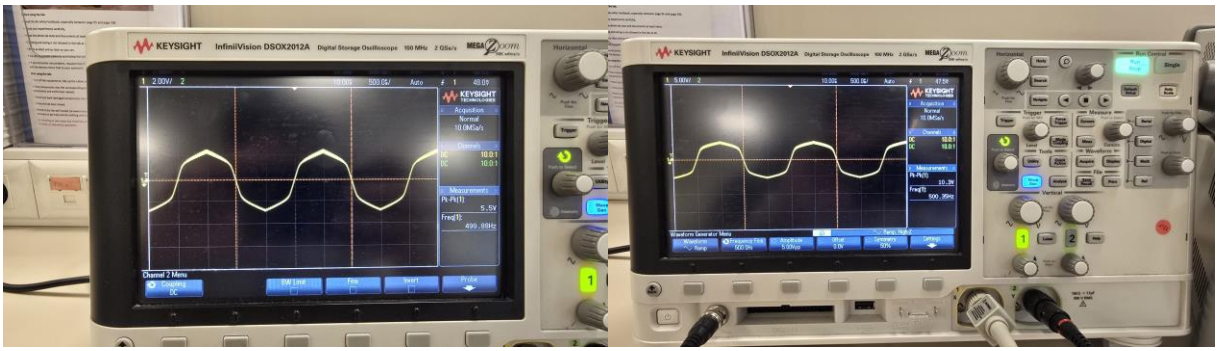


Figure 19. Sine wave generator at 500Hz with 5 and 10 V.

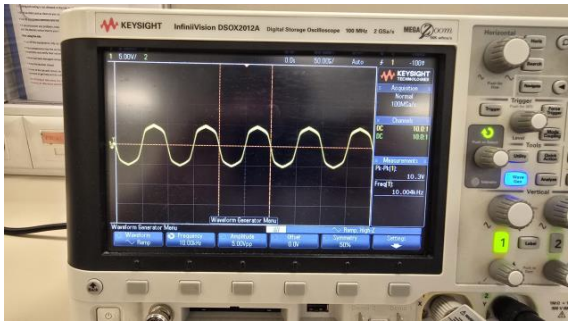


Figure 20. Sine wave generator at 10KHz with 10 V. (Since I forgot to take a photo of the 5V output, so I could not include it.)

In summary, this stage successfully demonstrated how a triangular waveform can be converted into a sine-like waveform through nonlinear piecewise-linear shaping. The resistor ladder created carefully chosen reference voltages, the diode branches introduced conduction at specific breakpoints, and the branch resistors controlled how much the slope was reduced in each region. The result was a rounded waveform with an initial measured amplitude of about 1.45 V. This signal was then amplified using an LM741 op-amp stage with appropriately selected resistor values so that the final output could meet the required 5 V and 10 V amplitude specifications. Therefore, both the theoretical calculations and the practical implementation agree well, and the triangular-to-sine wave generator block achieved its intended function in the experiment.

3.4 Cascaded System

In the final part of the experiment, the three main blocks of the function generator were connected together to form a complete cascaded system. These blocks were the Schmitt trigger, the integrator, and the triangular-to-sine shaping circuit, followed by an output amplification stage. Unlike the previous sections, where each block could be tested independently, in this part the circuit was operated as one complete system so that square, triangular, and sine-like waveforms could be generated from the same structure. The aim was to obtain output waveforms with amplitudes of 5 V and 10 V at both 500 Hz and 10 kHz, as required in the lab manual. The overall structure and operating idea of the cascaded system are also consistent with the pre-lab design, where the circuit was described as a feedback-based function generator composed of a Schmitt trigger, an integrator, a sine-shaping network, and a final amplification stage.

The system was implemented by first building the Schmitt trigger and integrator as a closed loop. This loop is the core of the oscillator, because it is responsible for continuously generating the square and triangular waveforms without needing a continuously varying external input. The Schmitt trigger works as a comparator with positive feedback. Its job is to switch its output sharply between two voltage levels whenever the input crosses certain threshold voltages. Because of the positive feedback, the switching thresholds are different for rising and falling input, which creates hysteresis and makes the transitions stable. In the implemented circuit, the triangular waveform coming from the integrator output was fed into the Schmitt trigger input. When this triangular waveform rose above the upper threshold, the Schmitt trigger output switched to one state; when it fell below the lower threshold, the output switched back to the opposite state. In this way, the Schmitt trigger continuously produced a square wave while monitoring the triangular signal from the integrator.

This square-wave output of the Schmitt trigger was then connected directly to the input of the integrator stage. The integrator was implemented using an LM741 op-amp in inverting configuration, with a resistor at the input and a capacitor in the feedback path. Its function is to integrate the constant high and low levels of the square wave over time. Since the input from the Schmitt trigger is approximately constant during each half-cycle, the integrator output changes linearly with time during those intervals. When the square-wave input is positive, the integrator output ramps in one direction, and when the square-wave input becomes negative, the integrator output ramps in the opposite direction. As a result, the output of the integrator becomes a triangular waveform. This triangular output is then fed back to the Schmitt trigger input, closing the loop. Therefore, the Schmitt trigger and integrator together form a self-

sustaining oscillation mechanism: the Schmitt trigger generates the square wave, the integrator turns that square wave into a triangular ramp, and the triangular ramp determines when the Schmitt trigger switches again. This is why the Schmitt trigger–integrator loop is the most important part of the cascaded function generator.

The frequency of oscillation in this loop is mainly determined by the integrator stage. As shown in the pre-lab, the integrator slope is given by **dVout/dt = -Vin / RC**

which means that the rate at which the triangular waveform rises or falls depends on the input voltage and on the RC time constant of the integrator. A larger RC value makes the ramp slower, resulting in a lower oscillation frequency, while a smaller RC value makes the ramp faster, resulting in a higher frequency. Based on the pre-lab analysis, the overall oscillation frequency of the cascaded system can be approximated as **f ≈ 1 / 4RC**

In this experiment, the capacitor value was kept constant at 47 nF, and the frequency was adjusted by changing the resistor value manually. Instead of using a potentiometer, standard resistor values were selected and replaced directly on the breadboard. This made the implementation simpler and followed the same design approach used in the pre-lab. For the low-frequency case, a resistor of 10 kΩ was used. Substituting this value into the oscillation formula gives **f ≈ 1 / [4(10 kΩ)(47 nF)] ≈ 500 Hz**.

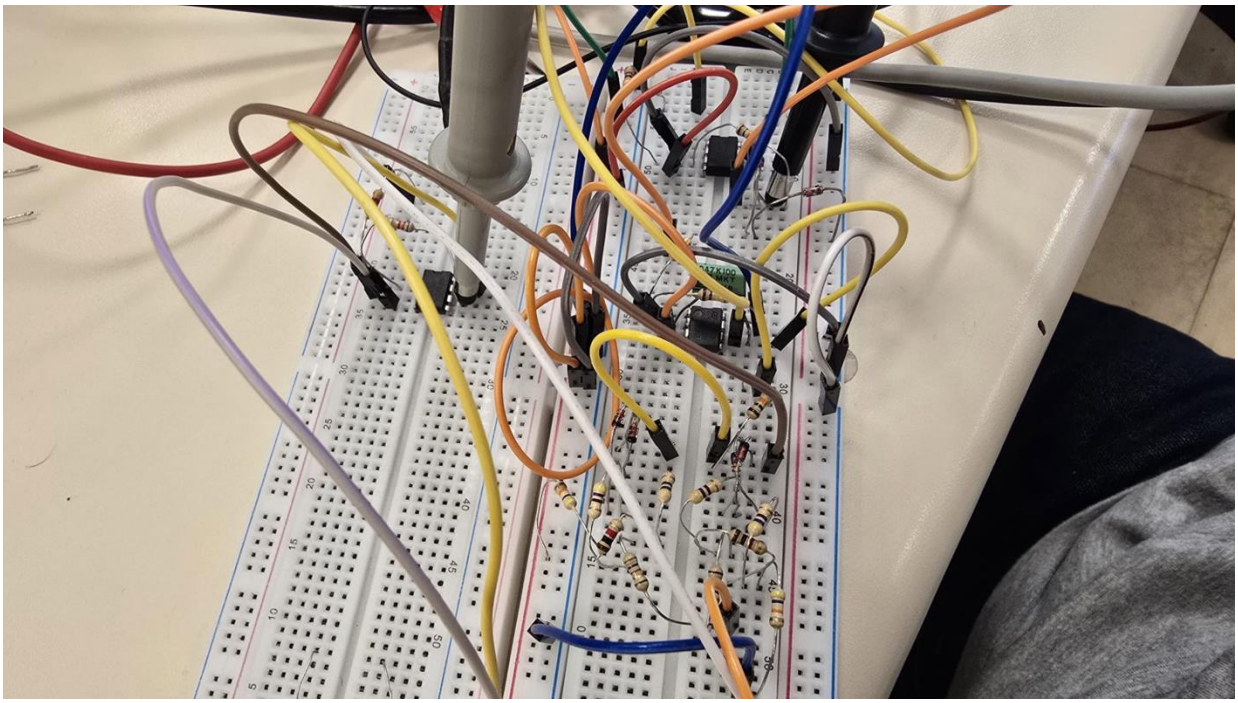


Figure 21. Breadboard implemenetatiton for 500 Hz output with 10 **kΩ resistor.**

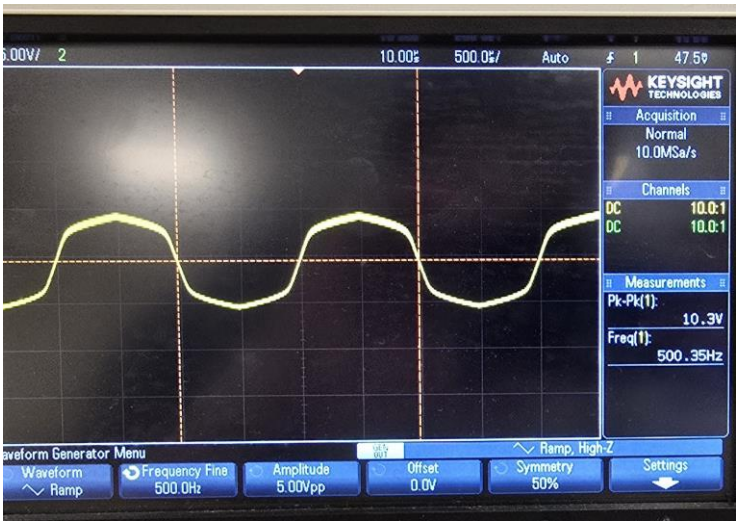


Figure 22. Output wave for 10V 0.5KJz

which matches the desired low-frequency operating point very well. For the high-frequency case, the resistor was reduced to 510 Ω. Using the same formula, **f ≈ 1 / [4(510 Ω)(47 nF)] ≈ 10.4 kHz**

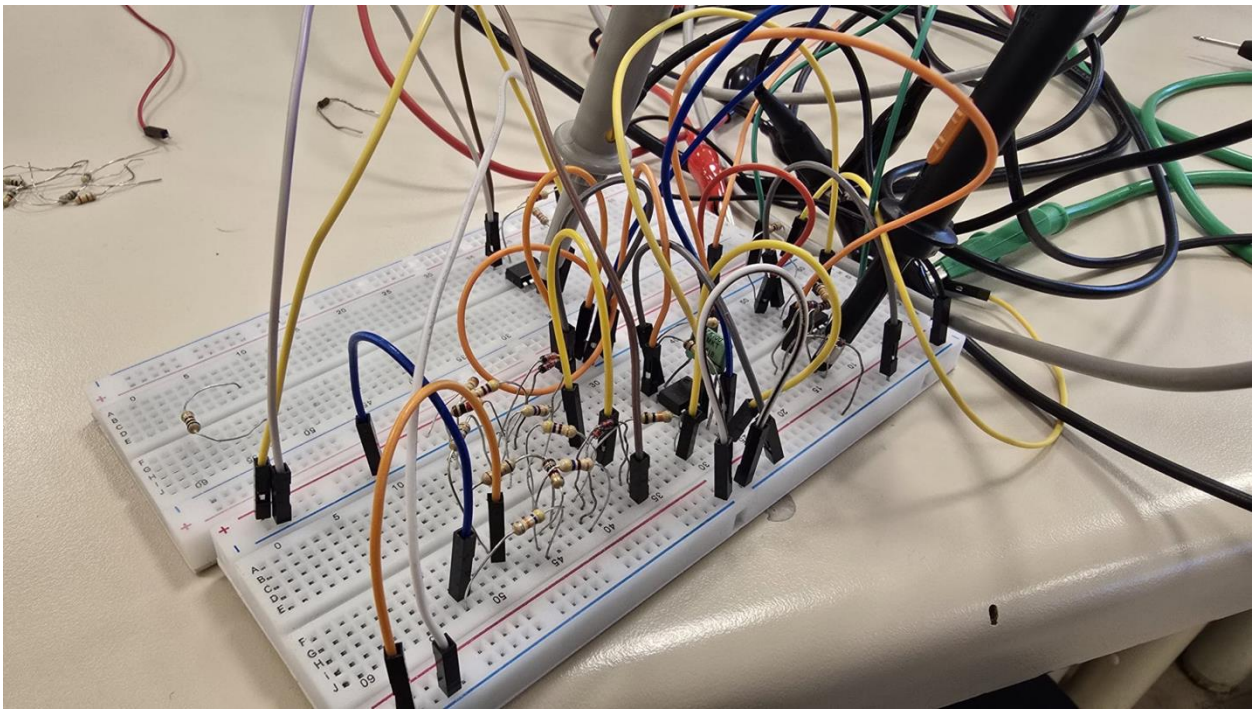


Figure 23. Breadboard implemenetatiton for 10KHz output with 510 **Ω resistor.**

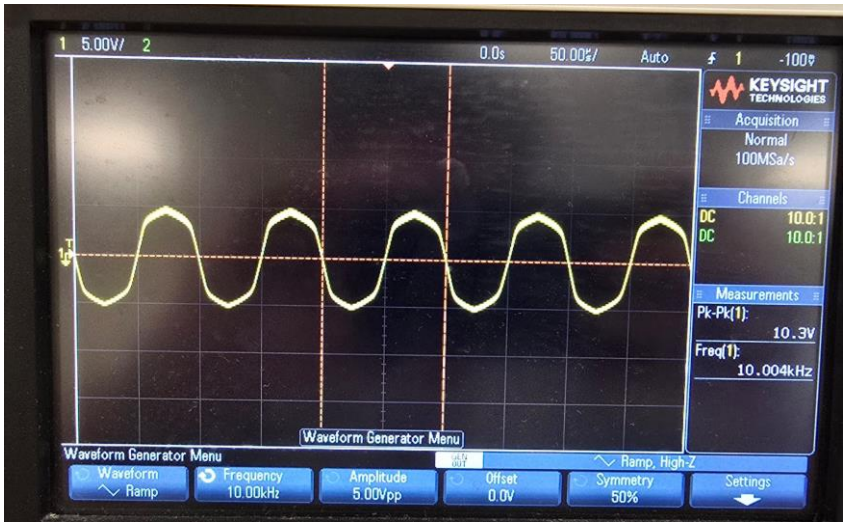


Figure 24. Output wave for 10V 10 KJz

which is very close to the target value of 10 kHz. Therefore, the resistor values of 10 kΩ and 510 Ω were chosen directly from the integrator-based oscillation formula, with the capacitor fixed at 47 nF. In practice, this means that the oscillation frequency was controlled by physically replacing the integrator input resistor on the breadboard while keeping the rest of the loop unchanged. This agrees with the pre-lab idea that frequency control in the cascaded system can be achieved by adjusting the integrator resistance while keeping the capacitor constant.

Once the Schmitt trigger–integrator loop was operating properly, two useful waveforms were already available in the system. The output of the Schmitt trigger provided the square wave, and the output of the integrator provided the triangular wave. The next step was to use the triangular waveform as the input to the sine-shaping stage. This stage was implemented using a nonlinear resistor-diode network. Its purpose was not to generate oscillation, but to reshape the already existing triangular waveform into a sine-like waveform. A triangular wave has a constant slope, which makes its corners sharp. A sine wave, however, has a gradually changing slope: it is steepest near the zero crossing and flatter near the peaks. The shaping network takes advantage of diode conduction at different voltage levels to reduce the slope of the triangular waveform progressively near its higher amplitudes. When the triangular input is small, the diodes are off and the output follows the input approximately linearly. As the input increases, diode branches begin to conduct one by one at their breakpoint voltages. Each conducting branch changes the effective resistance seen by the signal, reducing the slope in that region. Because different branches activate at different levels and the network is symmetric for positive and negative halves, the sharp triangular shape becomes rounded and approaches a sinusoidal waveform. This is why the stage is called a triangular-to-sine converter: it does not create a mathematically perfect sine wave, but it produces a much smoother, sine-like signal by piecewise-linear shaping.

In the pre-lab, this shaping behavior was explained by the breakpoint analysis of the resistor ladder and diode network. The ladder resistors create reference voltages symmetrically around zero, and when the diode forward drop is included, conduction starts at specific approximate levels. The pre-lab calculations showed that the effective breakpoint voltages are around 0.835 V and 2.19 V, with symmetric negative counterparts. This means that for small signal values the output remains almost unchanged, but after the first breakpoint the slope begins to decrease, and after the second breakpoint the waveform is

compressed even more strongly. Because the reduction happens gradually rather than abruptly, the triangular waveform is transformed into a smooth sine-like waveform instead of becoming clipped. This theoretical explanation was reflected in the experimental operation of the cascaded system, where the triangular output of the integrator was visibly rounded by the shaping stage.

After the sine-shaping stage, the resulting waveform still did not necessarily have the required output amplitude. The shaping circuit mainly changes waveform shape; it does not automatically produce the exact voltage levels demanded by the lab. For this reason, the final stage of the cascaded system was an op-amp amplifier used to adjust the output amplitude. In other words, after obtaining the sine-like waveform from the nonlinear shaping network, that signal was passed into an additional LM741 op-amp stage, and the gain of this stage was selected so that the final output could be set either to approximately 5 V or to approximately 10 V. This allowed the same shaped waveform to be scaled to the required amplitude without significantly changing its overall form. In practice, the amplifier stage was used as an output adjustment block: once the desired waveform shape had been obtained, the gain was tuned through resistor selection so that the final sine-like signal met the amplitude requirement. This is also consistent with the pre-lab description, where the last op-amp stage was used to scale the sine-like signal to the desired level.

The physical implementation of the cascaded system involved combining the three previously tested circuits on the breadboard. First, the Schmitt trigger and integrator were connected directly so that the Schmitt output drove the integrator input and the integrator output returned to the Schmitt input. After confirming that this loop oscillated correctly, the triangular output of the integrator was connected to the input of the sine-shaping network. The output of the shaping network was then connected to the final op-amp amplifier stage. The complete system therefore followed this signal path:

Schmitt trigger → Integrator → Sine-shaping circuit → Output amplifier

At the same time, the feedback path between the integrator and Schmitt trigger closed the oscillation loop. Thus, part of the system was responsible for waveform generation, and the later stages were responsible for waveform shaping and amplitude adjustment. The oscilloscope was used at different points in the circuit to verify the behavior of each stage: the Schmitt trigger output was checked for a square wave, the integrator output for a triangular wave, and the final output for a sine-like waveform at the desired amplitude.

When the resistor in the integrator was 10 kΩ, the system operated at about 500 Hz. Under this condition, the square, triangular, and sine-like waveforms were clearly observed, and the final output could be adjusted to approximately 5 V or 10 V depending on the gain setting of the last op-amp stage. When the resistor was changed to 510 Ω, the oscillation frequency increased to approximately 10 kHz, again in agreement with the integrator formula. At this higher frequency, the waveform still maintained its expected behavior, although some non-ideal effects such as slight distortion could be more noticeable due to op-amp and diode limitations. Even so, the system still generated the required waveforms successfully and demonstrated that the frequency could be controlled simply by changing the integrator resistor while keeping the capacitor fixed.

Overall, the cascaded system worked as intended and demonstrated the full function-generator principle. The Schmitt trigger and integrator formed the main oscillation loop and generated the square and triangular waveforms. The oscillation frequency was set by the RC time constant of the integrator,

and in this experiment it was controlled by fixing C = 47 nF and selecting R = 10 kΩ for approximately 500 Hz and R = 510 Ω for approximately 10 kHz. The triangular waveform was then sent to the sine-shaping network, where diode-based nonlinear conduction reduced the slope near the peaks and converted the signal into a sine-like waveform. Finally, this sine-like waveform was passed through an op-amp gain stage so that the output amplitude could be set to approximately 5 V or 10 V. In this way, all three blocks operated together as one complete cascaded function generator and successfully met the main requirements of the experiment.

4. Discussion/Analysis

4. Discussion / Analysis

In this experiment, the results obtained from theoretical analysis, simulation expectations, and experimental measurements were compared for each stage of the function generator system. Overall, the measured results were consistent with the expected theoretical behavior, although small deviations were observed, especially at higher frequencies.

A summary of the key results is given below:

Stage	**Condition**	**Expected**	**Measured**
Schmitt Trigger	500 Hz	~±5 V square	~5.3 Vpp
Schmitt Trigger	10 kHz	~±5 V square	~5.4 Vpp
Integrator	500 Hz	~5 V triangular	~5 V (ideal shape)
Integrator	10 kHz	reduced amplitude	~5 V with distortion
Sine Converter	500 Hz	sine-like	smooth sine-like
Sine Converter	10 kHz	sine-like	slightly distorted sine
Final Output	500 Hz	5 V / 10 V	achieved
Final Output	10 kHz	5 V / 10 V	achieved with minor distortion

Table 1.

From the Schmitt trigger results, the measured output closely matches the expected behavior. The output waveform was a clean square wave at both 500 Hz and 10 kHz, and the amplitude was successfully limited to around 5 V by the Zener diodes. The small difference between expected and measured values (around 0.3–0.4 V) can be explained by non-ideal Zener diode behavior and op-amp output limitations. In theory, the output is expected to be perfectly clipped, but in practice, diode tolerance and dynamic resistance slightly affect the exact voltage level.

For the integrator stage, the results at 500 Hz matched the theoretical predictions very well. The triangular waveform had a constant slope, and its amplitude was close to the expected value calculated from the relation dVout/dt = -Vin / (RC).
However, at 10 kHz, deviations from the ideal triangular shape were observed. The waveform showed distortions near the edges and small spikes. This behavior is expected because the LM741 op-amp has a limited slew rate and finite bandwidth. At higher frequencies, the op-amp cannot respond instantly to rapid changes in input, which causes non-linearities and distortion in the output waveform. Therefore, although the theoretical model assumes ideal integration, practical limitations of the op-amp become significant at high frequencies.

In the triangular-to-sine converter stage, the experimental results also followed the expected trend. At 500 Hz, the triangular waveform was successfully reshaped into a smooth sine-like waveform. This confirms that the diode-resistor network correctly reduced the slope near the peaks, as predicted by the breakpoint analysis in the pre-lab. The measured amplitude of approximately 1.45 V is also consistent with the expectation that this stage reshapes the signal rather than amplifies it.

At 10 kHz, the output still retained a sine-like shape but with noticeable distortion compared to the low-frequency case. This difference can be explained by two main factors. First, the diodes do not switch instantly; their finite switching speed affects the shaping accuracy at higher frequencies. Second, the op-amp used in the shaping and amplification stages has limited frequency response, which causes additional distortion. Therefore, while the theoretical model assumes ideal diode switching and perfect linear behavior between breakpoints, real components introduce frequency-dependent errors.

In the cascaded system, the overall behavior of the function generator was as expected. The Schmitt trigger and integrator formed a stable oscillation loop, and the frequency was successfully controlled by adjusting the resistor value in the integrator. The measured frequencies were close to the calculated values using $f \approx 1 / (4RC)$, which confirms the validity of the theoretical design. Using R = 10 kΩ produced approximately 500 Hz, and using R = 510 Ω produced approximately 10 kHz, as predicted.

The final sine-like output was successfully adjusted to 5 V and 10 V using the op-amp gain stage. The gain selection was based on the measured intermediate amplitude (~1.45 V), and the amplification stage performed as expected. However, slight deviations from ideal sine behavior were observed, especially at 10 kHz. These differences can be attributed to cumulative non-ideal effects from all stages, including op-amp limitations, diode nonlinearity, and breadboard parasitic capacitances and resistances.

In conclusion, the experimental results agree well with theoretical analysis and expected simulation behavior, particularly at lower frequencies. The main differences appear at higher frequencies, where real-world limitations of components become significant. These deviations are reasonable and expected, and they demonstrate the importance of considering non-ideal effects such as op-amp bandwidth, slew rate, and diode characteristics in practical circuit design.

5. Post-Lab

In this experiment, a complete function generator was designed, implemented, and analyzed using fundamental analog circuit building blocks. The main objective was to generate square, triangular, and sine-like waveforms at two different frequencies, 500 Hz and 10 kHz, and to adjust the output amplitude to approximately 5 V and 10 V. The overall system was constructed by cascading a Schmitt trigger, an integrator, a nonlinear shaping circuit, and a final amplification stage. Understanding the operation of this system requires examining both the individual behavior of each block and how they interact within the feedback loop.

A function generator essentially relies on transforming signals step by step. The key idea is to start with a signal that can be easily controlled, such as a square wave, and then process it through different circuit blocks to obtain other waveform shapes. In this design, the Schmitt trigger and integrator form the core of the system, creating a self-sustaining oscillation. The Schmitt trigger acts as a comparator with hysteresis, meaning that it switches its output between two voltage levels depending on whether the input signal crosses certain threshold values. These thresholds are not the same for increasing and decreasing input, which introduces stability and prevents noise from causing unwanted switching.

In the implemented circuit, the output of the Schmitt trigger was limited using Zener diodes to approximately ±5 V. This ensured that the amplitude of the square wave remained within a controlled range. The triangular waveform generated by the integrator was fed into the Schmitt trigger input. When the triangular waveform exceeded the upper threshold, the output of the Schmitt trigger switched to a negative value, and when it dropped below the lower threshold, it switched back to a positive value. This continuous switching created a square wave at the output of the Schmitt trigger. The presence of hysteresis is critical in this process because it ensures clean transitions and stable oscillation.

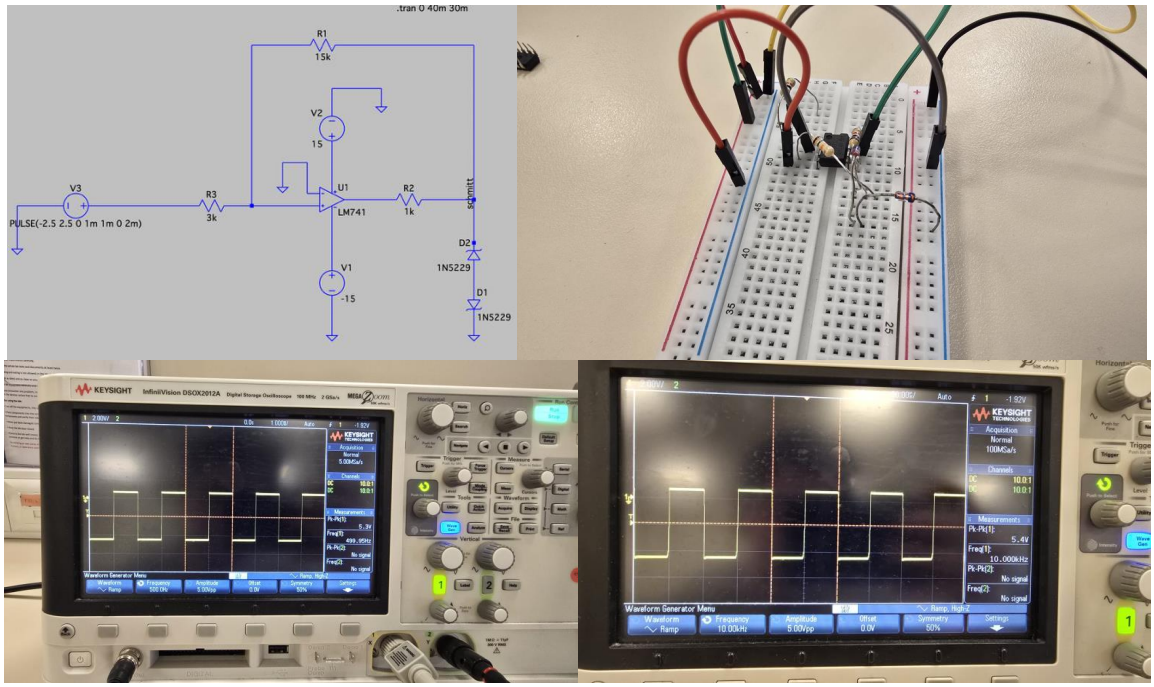


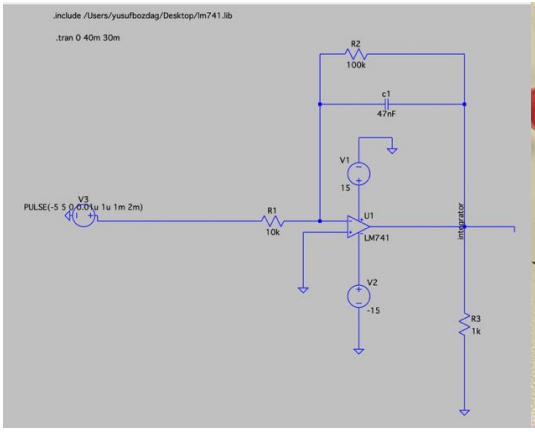
Figure 25. Important images for schmitt trigger

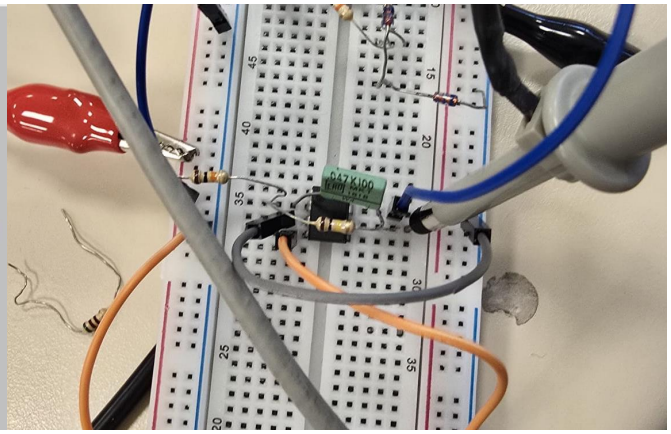
The output of the Schmitt trigger was then applied to the integrator stage. The integrator is an inverting op-amp configuration with a resistor at the input and a capacitor in the feedback path. Its function is to integrate the input signal over time. Since the input from the Schmitt trigger is a square wave with approximately constant positive or negative values during each half-cycle, the integrator produces a linear ramp during those intervals. This results in a triangular waveform at the output. When the input is positive, the output decreases linearly, and when the input is negative, the output increases linearly. The triangular waveform is then fed back to the Schmitt trigger, completing the loop.

The behavior of the integrator is governed by the equation dVout/dt = -Vin / (RC).
This equation shows that the slope of the triangular waveform depends on the input voltage and the RC time constant. A larger RC value results in a slower slope and lower frequency, while a smaller RC value produces a steeper slope and higher frequency. In this experiment, the capacitor value was fixed at 47 nF, and the frequency was adjusted by selecting different resistor values manually. For the low-frequency case, a resistor of 10 kΩ was used. This gives RC = 470 µs. For a 500 Hz signal, the period is 2 ms, and half of the period is 1 ms. Using the equation, the voltage change over half a period is approximately ΔV ≈ (5 V / 470 µs) × 1 ms ≈ 10.6 V, which corresponds to a triangular waveform with an amplitude close to ±5 V. The measured waveform at 500 Hz matched this expectation very well, showing a clean and symmetric triangular shape.

For the high-frequency case, the resistor was reduced to 510 Ω while keeping the capacitor constant. This significantly reduces the RC time constant, increasing the slope of the waveform. The oscillation frequency can be approximated by f ≈ 1 / (4RC).

Substituting the values gives a frequency close to 10 kHz, which matches the target value. However, at this higher frequency, the measured triangular waveform showed distortions, particularly near the corners. These distortions are mainly due to the limitations of the LM741 op-amp. The op-amp has a finite slew rate, meaning that it cannot change its output voltage infinitely fast. At higher frequencies, the required rate of change exceeds the op-amp's capability, leading to rounding and distortion of the waveform. Additionally, the limited bandwidth of the op-amp affects its ability to accurately reproduce high-frequency signals.





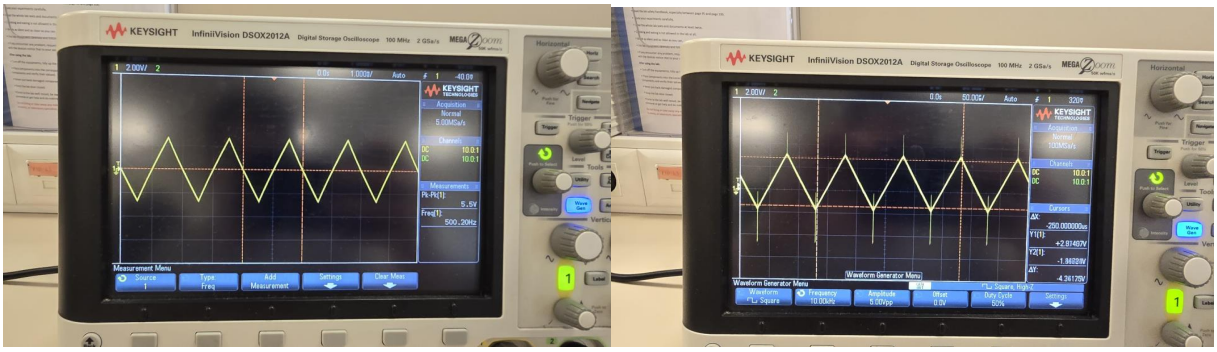
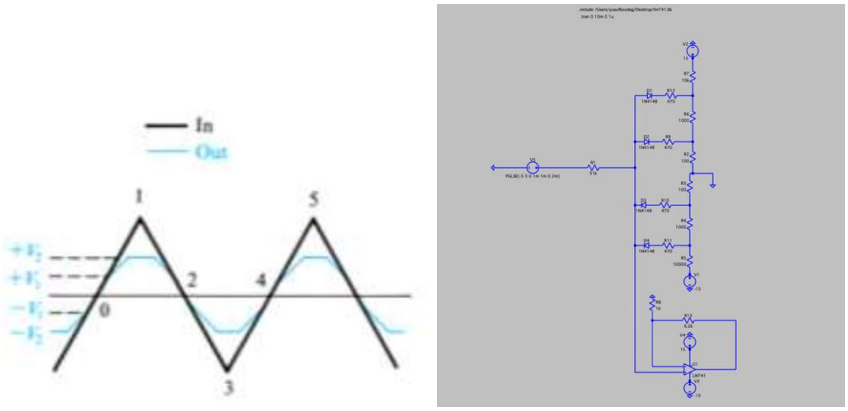


Figure 26. Important images for integrator

Once the triangular waveform was obtained, it was passed through the triangular-to-sine shaping circuit. This circuit is based on a nonlinear resistor-diode network and is designed to gradually modify the shape of the triangular waveform. A triangular wave has a constant slope, while a sine wave has a varying slope that is maximum at the zero crossing and minimum at the peaks. The shaping circuit achieves this transformation by reducing the slope of the waveform as the voltage increases. At small signal levels, the diodes are off, and the circuit behaves linearly, allowing the signal to pass unchanged. As the voltage increases, the diodes begin to conduct at specific breakpoint voltages determined by the resistor ladder.

These breakpoint voltages were calculated in the pre-lab analysis. By considering the voltage division in the resistor ladder and the forward voltage drop of the diodes (approximately 0.7 V), the breakpoints were found to be around 0.835 V and 2.19 V. When the input signal exceeds these values, additional current paths are created through the diodes, effectively reducing the slope of the waveform. This process occurs symmetrically for both positive and negative halves of the signal, resulting in a smoother waveform that resembles a sine wave. The measured output of this stage had an amplitude of approximately 1.45 V, indicating that the circuit reshapes the waveform without significantly amplifying it.



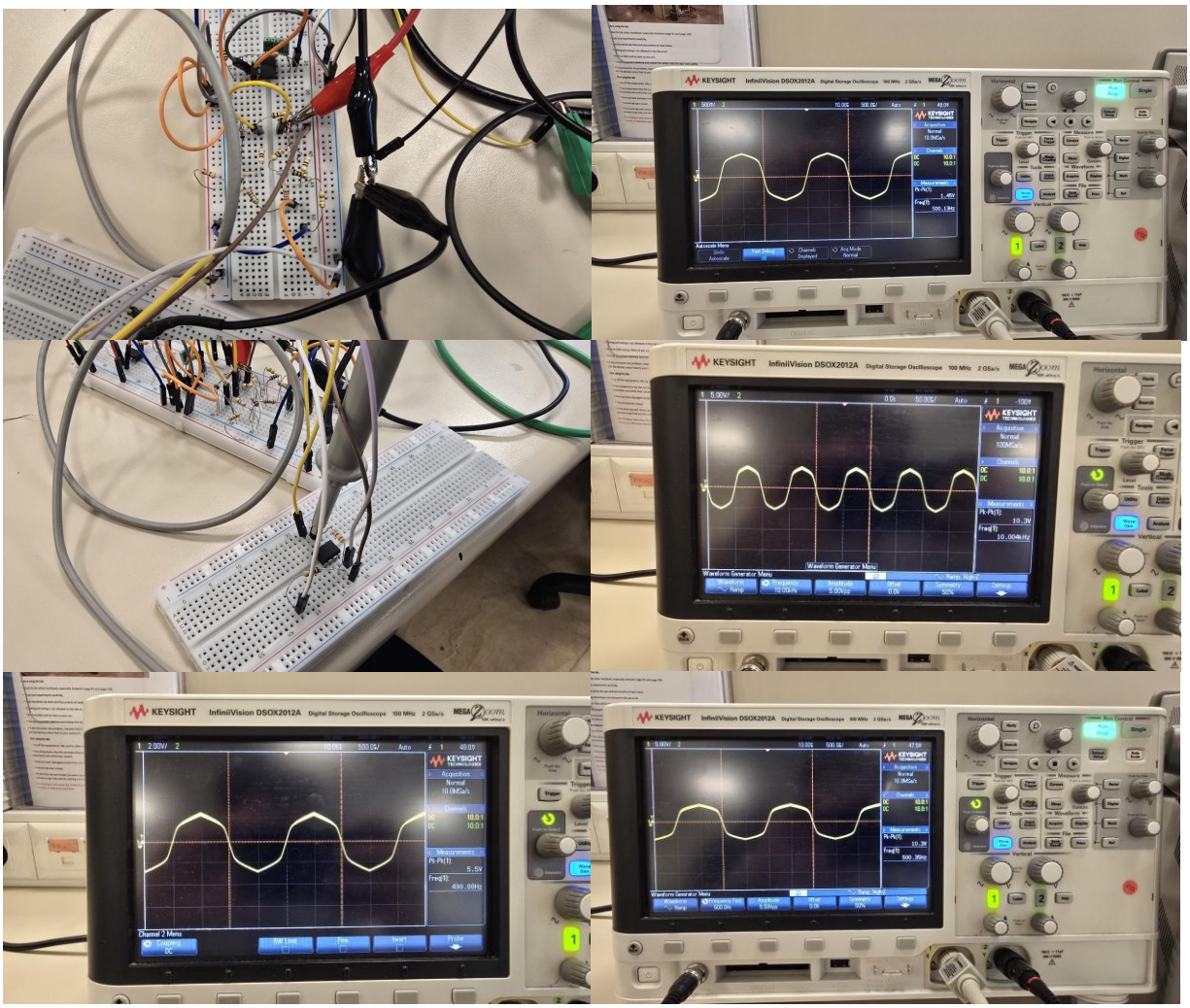


Figure 27. Important images for sine wave generator

At 500 Hz, the output of the shaping circuit closely resembled a sine wave, with smooth transitions and rounded peaks. This confirms that the diode-resistor network was functioning as intended. At 10 kHz, however, the waveform showed noticeable distortion. This can be explained by the finite switching speed of the diodes and the frequency limitations of the op-amp. At higher frequencies, the diodes do not turn on and off instantaneously, and the op-amp cannot maintain ideal linear behavior, leading to deviations from the expected shape.

To achieve the required output amplitudes of 5 V and 10 V, the sine-like signal was then passed through an additional op-amp amplification stage. The gain of this stage was determined using the formula $A_v = 1 + R_f / R_{in}$. By selecting appropriate resistor values, the amplitude of the signal was scaled from approximately 1.45 V to the desired levels. This stage allowed precise control of the output amplitude without significantly altering the waveform shape. The final outputs at both 500 Hz and 10 kHz were successfully adjusted to approximately 5 V and 10 V, although some distortion remained at the higher frequency.

In the cascaded configuration, the Schmitt trigger and integrator were connected in a feedback loop to create a self-sustaining oscillation. In this loop, the square wave produced by the Schmitt trigger is continuously fed into the integrator, while the triangular output of the integrator is fed back to the Schmitt trigger input. This interaction allows the system to oscillate without requiring an external time-varying input. The oscillation frequency is mainly determined by the RC time constant of the integrator. In this experiment, the capacitor value was kept constant at 47 nF, and the frequency was controlled by manually changing the resistor value. By selecting R = 10 kΩ, an oscillation frequency of approximately 500 Hz was obtained, while reducing the resistor to R = 510 Ω increased the frequency to around 10 kHz. These results are consistent with the theoretical relation f ≈ 1 / (4RC), confirming that the Schmitt trigger–integrator loop successfully determines the operating frequency of the system.

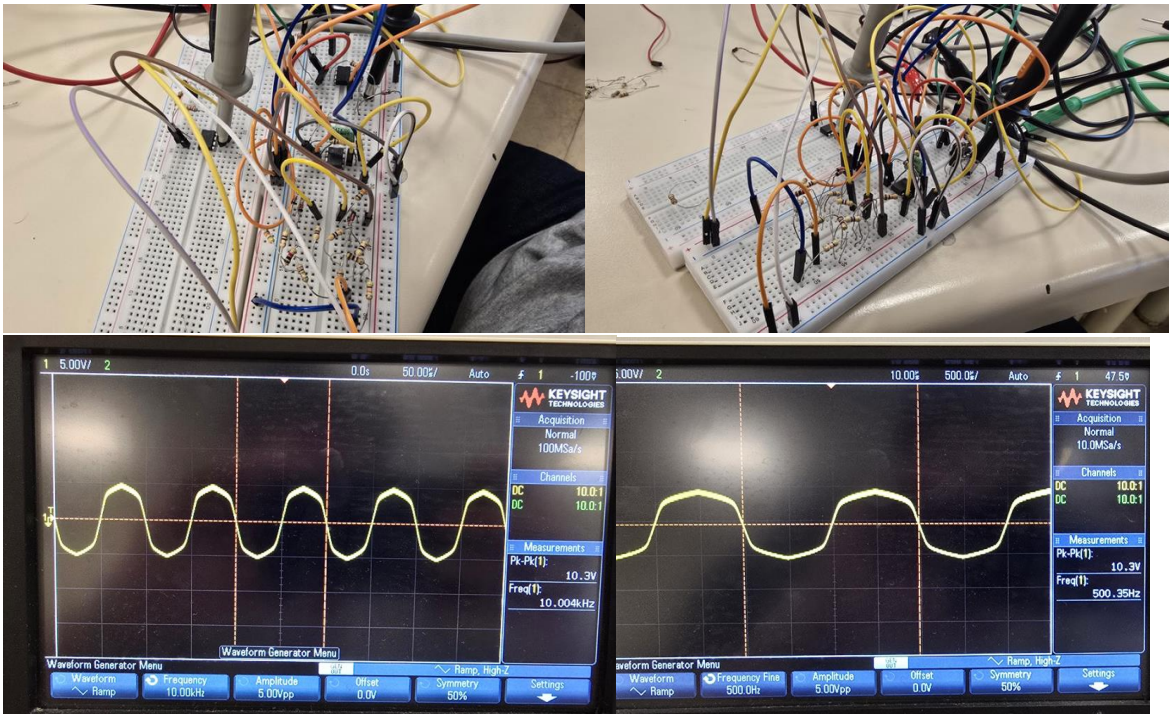


Figure 28. Important images for cascaded system

When comparing theoretical calculations with experimental results, it is clear that the system behaves as expected at lower frequencies. The measured waveforms closely match the theoretical predictions, and the calculated frequencies and amplitudes are consistent with the observed values. At higher frequencies, however, discrepancies become more significant. These differences are mainly due to non-ideal characteristics of the components used in the circuit. The LM741 op-amp has limited slew rate and bandwidth, which affects its performance at high frequencies. The diodes also have non-ideal switching behavior, and the breadboard introduces parasitic capacitances and resistances that are not accounted for in the theoretical model.

Despite these limitations, the overall operation of the function generator was successful. The Schmitt trigger provided stable switching behavior, the integrator controlled the frequency of oscillation, the shaping circuit converted the triangular waveform into a sine-like waveform, and the amplification stage

adjusted the output amplitude. The experiment demonstrated how complex signal generation can be achieved by combining simple analog circuit blocks and highlighted the importance of understanding both theoretical principles and practical limitations.

In conclusion, the experiment provided a comprehensive understanding of function generator design and operation. It showed how different waveform types can be generated and controlled using fundamental circuit elements and how real-world factors influence circuit performance. The results confirm that the theoretical design is valid, especially at lower frequencies, while also emphasizing the need to consider non-ideal effects in practical implementations.

6. Conclusion

In this experiment, a complete function generator was successfully designed, implemented, and analyzed using fundamental analog circuit components. The main objective was to generate square, triangular, and sine-like waveforms at specific frequencies (500 Hz and 10 kHz) and to control their amplitudes at approximately 5 V and 10 V. Based on the results obtained from theoretical calculations, circuit implementation, and experimental measurements, it can be concluded that the overall design approach was valid and effective, especially at lower frequencies.

The results confirm that the Schmitt trigger and integrator combination provides a reliable method for generating oscillations. The Schmitt trigger produced a stable square wave with clearly defined switching levels, while the integrator successfully converted this signal into a triangular waveform. The frequency of oscillation was accurately controlled by the RC time constant of the integrator, and the experimental results closely matched the theoretical prediction given by $f \approx 1 / (4RC)$. By selecting R = 10 kΩ and R = 510 Ω while keeping the capacitor fixed at 47 nF, oscillation frequencies of approximately 500 Hz and 10 kHz were obtained, validating the design calculations.

The triangular-to-sine shaping circuit also performed as expected. At lower frequencies, the diode-resistor network effectively reduced the slope of the triangular waveform near its peaks, resulting in a smooth sine-like output. This demonstrates that the piecewise-linear shaping method used in the design is a practical approach for approximating sinusoidal signals using simple components. However, the results also show that the quality of the sine wave depends on frequency. At 10 kHz, the output waveform exhibited noticeable distortion, indicating that the shaping process becomes less accurate as frequency increases.

The amplification stage successfully scaled the output signal to the required amplitude levels. By selecting appropriate resistor values in the op-amp configuration, the sine-like waveform was adjusted to approximately 5 V and 10 V outputs. This confirms that amplitude control can be achieved independently of waveform generation, which is an important design feature of function generators.

Although the overall behavior of the system matches theoretical expectations, some deviations were observed in the experimental results, particularly at higher frequencies. These differences can be attributed to several non-ideal factors. The LM741 op-amp has limited slew rate and bandwidth, which restrict its ability to respond to rapid signal changes at high frequencies. As a result, waveform distortions appear in both the integrator and shaping stages. Additionally, the diodes used in the shaping circuit do not switch instantaneously, and their forward voltage characteristics introduce small variations

in the breakpoint behavior. Parasitic effects from the breadboard, such as stray capacitance and resistance, also contribute to discrepancies between theoretical and measured results.

Despite these limitations, the experiment clearly demonstrates the fundamental operating principles of a function generator. The ability to generate different waveform types through sequential signal transformations highlights the importance of understanding how basic circuit blocks interact within a system. Furthermore, the results support the hypothesis that a cascaded analog system consisting of a Schmitt trigger, integrator, nonlinear shaping network, and amplifier can effectively produce controlled waveforms with adjustable frequency and amplitude.

In conclusion, the experiment was successful in achieving its objectives and provided valuable insight into both theoretical and practical aspects of analog circuit design. It reinforced the relationship between mathematical models and real-world implementations while also emphasizing the impact of component limitations on circuit performance. These findings are important for understanding more advanced signal generation systems and for developing more accurate and efficient circuit designs in future applications.

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